

Midterms Skills Exam - Gr2

Course Code: CPE 411	Program: Computer Engineering
Course Title: Interactive Extended Reality	Date Performed: March 14, 2026
Section: CPE 411-CPE22S4	Date Submitted: March 14, 2026
Name(s): Bulaong, Ross Andrew D. Esguerra, Serge Palomares, Charles Yexel Reyes, Jamie Rose	Instructor: Engr. Jimlord M. Quejado

1. Discussion

Make an app that detects flat surfaces and allows the user to spawn a portal on their floor or wall. Then complete a challenge to create an interactive character app.

2. Materials and Equipment

Unity Hub and Unity Editor
Learn Unity Website ([Unity Learn](#))
Computer with Network Connection

3. Output

BULAONG

The screenshot displays two side-by-side browser windows from the Unity Learn platform.

Left Window (Profile Page):

- Header:** Shows the user's name "BULAONG, ROSS ANDREW" and a "Logout" button.
- Left Sidebar:** Contains navigation links: Account, Dashboard, Courses, Groups, Calendar, Inbox, History, and Help.
- Main Content Area:**
 - Notifications:** Includes links for Profile, Files, Settings, ePortfolios (Legacy), Portfolio, QR for Mobile Login, and Global Announcements.
 - Accessibility Settings:** Shows options for "Use High Contrast UI" (checked) and "Use a Dyslexia Friendly Font" (unchecked).
 - Skills Exam:** A section titled "Skills Exam: Create a Plane AR App" with a description: "detects flat surfaces and allows the user to spawn a portal on their floor or wall. Then complete a challenge to create an interactive character app." It also mentions a mission in the Unity pathway: "Creating an AR Application. Interact with something else and demonstrate its use. Complete this challenge through a short 10-second video. Watch the short video and screenshots in a lab report."
 - Lifelong Learning:** A section with a "Rating" table:

6 pts	5 pts	4 pts
Excellent Educational interests and pursuits	Good Educational interests and pursuits	Satisfactory Educational interests and pursuits

Right Window (Results Page):

- Header:** "Your results"
- Message:** "Congratulations! You passed. If you would like to score higher, you can [retake](#) the quiz. We always save your highest score."
- Score:** "Passed • You got 9 of 10 correct" with a progress bar.
- XP Gains:**
 - +20 XP: You gained 20 XP points towards Art & Animation
 - +20 XP: You gained 20 XP points towards Systems Design
 - +20 XP: You gained 20 XP points towards Programming
- Completion Date:** "You completed this quiz on March 14, 2026"
- Next Step:** "Next up in Mobile AR Development" with a badge for "AR experience design" (+300 XP, 9 Hours 45 Mins).
- Buttons:** "Continue to next"
- Footer:** Unity logo, language options (English, Deutsch, 日本語, Français, Português, 简体中文, Español, Русский, 한국어), and social media links.

Midterm Skills Exam: Create a Plane Detection AR App

Shows to watch

School Shit

My Loving Girlfriend

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Global Announcements

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Use a Dyslexia Friendly Font

Immersive Reality

Detection AR App

Skills Exam: Create a Plane Detection AR App

1:59p.m. Points 24

upload File Types pdf

far 5 at 12a.m.

detects flat surfaces and allows the user to their floor or wall. Then complete a challenge active character app.

e given mission in the unity pathway: Creating and AR Application.

ter to something else and demonstrate its use. through a short 10-second video.

the short video and screenshots in a lab report.

Lifelong Learning

Rating

6 pts Excellent | Educational interests and pursuits

5 pts Good | Educational interests and pursuits

4 pts Satisfactory | Educational interests and pursuits

Shows to watch

Midterm Skills Exam - Google Docs

Mobile AR Development

All Bookmarks

Download pathway outline csv

Mobile AR Development v2022.3 Common Cartridge File

Content Outline

Expand all

Unity Version 2022.3

77% complete

MISSION

Get started with AR

MISSION • COMPLETED MARCH 14, 2026

Create a basic face filter

MISSION • COMPLETED MARCH 05, 2026

Create an interactive face filter

MISSION • COMPLETED MARCH 07, 2026

Create a marker-based AR app

MISSION • COMPLETED MARCH 14, 2026

Create a plane detection AR app

MISSION

AR experience design

ESGUERRA

Calendar

tip...

Incognito (2)

Relaunch to update

Facebook - Log In o...

医院建筑设计模型 ...

All Bookmarks

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ESGUERRA, SERGE

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Use a Dyslexia Friendly Font

PATHWAY

Mobile AR Development

Unity Version: 2022.3

Change

Contents

TUTORIAL

Limit the visibility of the portal world

EXERCISE

Submission: AR friend

QUIZ

Create with AR: markers and planes Quiz

1 • Question 1

2 • Question 2

3 • Question 3

Your results

Congratulations! You passed. If you would like to score higher, you can [retake](#) the quiz. We always save your highest score.

Passed • You got 10 of 10 correct

+20 XP

You gained 20 XP points towards Art & Animation

+20 XP

You gained 20 XP points towards Systems Design

+20 XP

You gained 20 XP points towards Programming

You completed this quiz on March 14, 2026

Next up in Mobile AR Development

6

AR experience design

+300 XP • 9 Hours 45 Mins

Continue to next

Activate Windows

Go to Settings to activate Windows.

Calendar

tip... Incognito (2) Relaunch to update

Facebook - Log In... 医院建筑设计模型... All Bookmarks

ESGUERRA, SERGE

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PATHWAY

Mobile AR Development

Unity Version: 2022.3

Contents

MISSION

Create a plane detection AR app

TUTORIAL

Set up a scene with plane detection

TUTORIAL

Detect the user's touch

TUTORIAL

Detect when the user touches a plane

TUTORIAL

Instantiate the portal on a plane

TUTORIAL

Test the frame rate of the portal world

MISSION

Create a plane detection AR app

Make an app that detects flat surfaces and allows the user to spawn a portal on their floor or wall. Then you'll create an interactive character app.

8 Mins

What you will learn

In this project, you'll create an AR app that makes use of plane detection technology. iOS and Android devices have the ability to detect flat surfaces — or planes — in your physical environment. This includes horizontal planes like the floor or a table, and vertical planes like the walls.

Additional XP earned by completing the unit

+300

You completed this and earned the XP on Mar 14, 2026

You will create an app where the user will be able to tap on a plane

MADRIAGO

(9) Fac... Midt... Midt... Create a plane detection AR app

tip.instructure.com/course... learn.university.com/pathway/mobile-ar-development/unit/create-a-plane-detection-ar-app?version=2022.3

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MADRIAGO, DAVE ADRIANE

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Test the frame rate of the portal world

TUTORIAL

Limit the visibility of the portal world

EXERCISE

Submission: AR friend

What you will learn

In this project, you'll create an AR app that makes use of plane detection technology. iOS and Android devices have the ability to detect flat surfaces — or planes — in your physical environment. This includes horizontal planes like the floor or a table, and vertical planes like the walls.

You will create an app where the user will be able to tap on a plane in order to place a portal onto it. This portal will be a window into a virtual 3D environment that the user can peer into and explore.

You will also complete a challenge where you design an app that allows the user to place a character on a table and interact with it.

Skills you will learn

Absolute Beginner AR Design

Absolute Beginner Visual Scripting

Absolute Beginner AR Development

Additional XP earned by completing the unit

+300

Unity

LANGUAGE

English Deutsch 日本語 Français Português 简体中文 Español Русский 한국어

SOCIAL

in

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o

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Midterm Skills Exam: Creat...

tip.instructure.com/courses/72332/assignments/2953495

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Skills Exam: Create a Plane Detection AR App

Start Assignment

1:59pm Points 24 Submitting a file upload Available after Mar 5 at 12am

detects flat surfaces and allows the user to spawn a portal on their floor to complete a challenge to create an interactive character app.

the given mission in the unity pathway: Creating a Marker-based AR

ter to something else and demonstrate its use. Demonstrate through a video, the short video and screenshots in a lab report.

Lifelong Learning

				Ratings
6 pts	5 pts	4 pts	3 pts	
Excellent Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge learned into practice	Good Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge independently	Unsatisfactory Beginner level knowledge	
6 pts	5 pts	4 pts	3 pts	
Excellent Completes an	Good Completes an	Satisfactory Requires	Unsatisfactory Requires	

PATHWAY Mobile AR Development

Unity Version: 2022.3 Change

Contents

QUIZ Create with AR: markers and planes Quiz

1 • Question 1

2 • Question 2

3 • Question 3

4 • Question 4

5 • Question 5

6 • Question 6

7 • Question 7

Involving AR markers and AR detection planes.

Retake Quiz

Review the questions you missed and retake the quiz

Attempted • You got 7 of 10 correct 8 of 10 to pass

Retake in 23h 50m

Question 1

You're creating a marker-based AR Museum app that will show specific objects at life size in the user's environment.

One of the art pieces is a sculpture that has a real world height of 2 meters. How many Unity units tall should the 3D object be? Select one answer

PALOMARES

tip.instructure.com

TIP

PALOMARES, CHARLES YEXEL

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PATHWAY Mobile AR Development

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+20 XP You gained 20 XP points towards Art & Animation

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You completed this quiz on March 14, 2026

Next up in Mobile AR Development

AR experience design +300 XP • 9 Hours 45 Mins

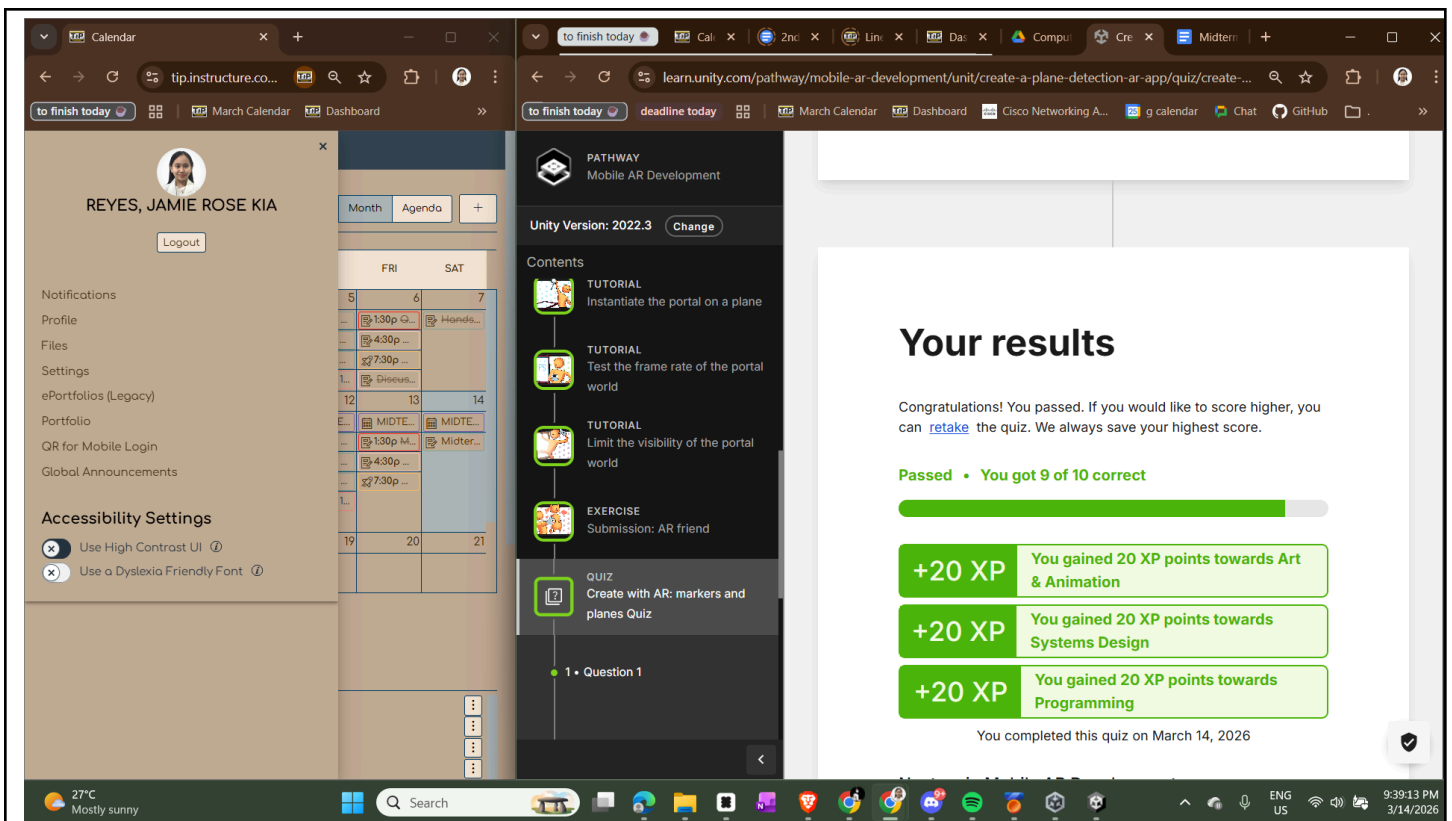
Continue to next

24°C Partly cloudy

Search

ENG US

9:47 pm 14/03/2026



4. Conclusion

Bulaong, Ross Andrew D.

The AR plane detection mission proved to be both difficult and educational. My initial work could not correctly identify object positions, but I pushed through it. The errors I encountered during the testing process helped me understand how augmented reality works with actual physical environments. The experience developed my technical abilities while making me prepared for upcoming AR projects which will involve more advanced work.

Esguerra, Serge

In conclusion, this mission taught me how to develop a plane detection AR app. During the process, I faced some challenges such as the app failing to detect planes and objects appearing with incorrect rotation. Despite these difficulties, I learned how to fix them and made the app work properly. In addition, I also tried to customize the app by changing the model with the model for our final project, which is the laboratory room. This experience not only improved my technical skills but also enhanced my problem-solving abilities. Overall, it gave me the confidence to apply AR technology creatively in future projects.

Palomares, Charles Yexel

This activity (Midterm Skill Exam) teaches me how to build a plane detection AR app. I have a lot of trouble because the plane does not show and the model rotates the wrong way. But I fixed all the problems and changed the model to my computer table. This improves my technical skill and helps me learn how to solve problems better. Now I have big confidence to use AR technology for more creative projects later. Thus, I will try to practice more to be able to create my own AR project in the future.

Madriago, Dave Adriene

In conclusion, completing this mission is very challenging because of its confusing concepts, but it's very enjoyable at the same time. Learning how to detect planes and place objects through touch helped me understand how digital objects can appear in the real world. This work helped my skills and also push my learning capabilities, it helped me understand how AR works in the actual real life.

Reyes, Jamie Rose

In conclusion, completing this mission on how to create a detection AR app was both challenging and fun for me. At first, the ideas were a little confusing, but as I learned how to detect physical planes and place objects using touch, I started to understand how digital objects can appear in the real world. Creating the portal and controlling when it appears also helped me understand how AR experiences are made while still thinking about mobile performance. I believe that the

